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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

SE 19-206-0602 PROCESS INSTRUMENTATION AND CONTROL (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain the measurement principles and techniques of measuring temperature and pressure.

CO2: Compare the methods for the measurement of flow, level, pH and humidity.

CO3: Interpret the operation of different types of control system.

CO4: Analyse the stability of a control system.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Explain about drifts in measurement.	3	3	L1	1	1
(b) Give a note on errors in measurement.	3	3	L1	1	3
(c) List the advantages of Pitot meter.	3	3	L2	2	2
(d) Define pH. Explain any method to measure pH.	3	3	L2	2	2
(e) Compare Gain margin and Phase Margin.	3	3	L3	3	2
(f) Explain the characteristics of PI controller.	3	3	L2	3	3
(g) Briefly describe about Error Detectors.	3	3	L2	4	1
(h) Explain Delay time and Rise time.	3	3	L1	4	1

PART B

(4 × 12 = 48)

II. (a) Detailed description about construction and working of Resistant Thermometers.	5	5	L2	1	1
(b) Draw the flowchart for elements of measurements.	2	2	L3	1	2
(c) Explain the principle and working procedure of Optical Pyrometer.	5	5	L2	1	2
OR					
III. Discuss about different types of temperature measuring instruments. with neat figures. Also describe their industrial applications.	12	12	L2	1	3
OR					
IV. (a) Differentiate Flumes and Weirs.	7	7	L3	2	2
(b) Elucidate about any variable area flow meters.	5	5	L3	2	2
OR					
V. (a) Explain about any two flow measuring devices. Compare its merits and demerits.	10	10	L2	2	1
(b) Give a note on application of Venturi meter.	2	2	L1	2	1

(P.T.O.)

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		Marks	BL	CO	PO
VI.	(a) What is a second order system? Give its general form.	6	L1	3	1,2
	(b) Describe a simple thermal system in which incoming liquid is heated by the heater in the tank and going out with a higher temperature. Develop first order transfer function of this thermal process.	6	L2	3	1,2
OR					
VII.	(a) Describe about time domain for a transient system.	6	L2	3	1
	(b) Distinguish PI, PD and PID controllers.	6	L3	3	1
VIII.	(a) Describe Routh-Hurwitz stability criterion.	6	L1	4	1
	(b) What is controller tuning?	6	L1	4	2
OR					
IX.	(a) Elucidate on Control System Components.	6	L2	4	1
	(b) Explain Nyquist stability criterion.	6	L2	4	2

Bloom's Taxonomy Levels

L1 – 24.16 %, L2 – 56.66%, L3 – 19.16%.
